ISTIAK AHMED

SOFTWARE ENGINEER

Columbia, Missouri / (573) 673-5513 / istiak1021@gmail.com / LinkedIn / Google Scholar / Portfolio

Software engineer with **4+ years building Unity and C#** across immersive AR, VR and MR experiences, live mobile titles and high-performance WebGL games, developing hands-on for Meta Quest, HoloLens, Magic Leap and HTC Vive. That work is grounded in event-driven C# architecture, multiplayer networking, in-headset UI and performance optimization for standalone builds. Currently developing real-time XR systems and in-headset interfaces at the University of Missouri, where the research has produced six peer-reviewed publications at top-tier XR venues.

TECHNICAL SKILLS

LANGUAGES	C#, C++, C, Python, Kotlin, Java
ENGINES & TOOLS	Unity (AR, VR, MR, WebGL, animation, game development), Unreal Engine (familiar), Android Studio (native Android in Kotlin and Java)
XR SDKS	Meta XR SDK, OpenXR, XR Interaction Toolkit, MRUK
HEADSETS	Meta Quest, Microsoft HoloLens, Magic Leap, HTC Vive
NETWORKING	Photon PUN 2, Photon Fusion, Unity Netcode for GameObjects, Mirror, Unity Relay & Lobby, PlayFab, RESTful API integration in Unity
ENGINEERING	Event-driven architecture, OOP and design patterns, Unity Profiler and build optimization, in-headset UI / HUDs, full-body and hand tracking, Git, Jira
MACHINE LEARNING	PyTorch, TensorFlow, LLMs, computer vision, reinforcement learning, explainable AI

EXPERIENCE

Graduate Research Assistant
JAN 2025 – PRESENT

Dependable Cyber-Physical Systems (DCPS) Laboratory, University of Missouri — Columbia, MO

TRUSTWORTHY XR: SECURITY, PRIVACY, AND AI-DRIVEN INTERACTION · FUNDED BY THE U.S. ARMY RESEARCH LABORATORY (ARL)

- Build Unity AR, VR and MR applications in C# for Meta Quest, HoloLens, Magic Leap and HTC Vive, from prototype through deployed standalone builds.
- Develop real-time systems — networked data pipelines, REST service integration and event-driven architecture, keeping heavy processing off the render thread.
- Implement in-headset UI and interaction flows, including calibration, input handling and guided user sessions.
- Profile and optimize builds to hold target frame rates; published six peer-reviewed papers at top-tier XR venues.

Software Engineer
OCT 2023 – NOV 2024

Torpedo Labs, Inc. — 218 Carmel Sky Street, Las Vegas · Remote

- Shipped features for a live casino-style slots title in Unity and C#, working in a mature production codebase against an active player base.
- Developed multiple distinct game types inside a single application, sharing one economy, state machine and UI framework across them.
- Debugged and resolved production issues on live builds, owning fixes end to end from reproduction through release.
- Refactored legacy systems for reusability and maintainability, enforcing consistent design patterns and clean-code standards in review.

Unity Developer
NOV 2021 – SEP 2023

Softwind Tech
Dhaka, Bangladesh

- Built immersive VR experiences for enterprise and telecom clients, including a large-scale Hajj simulation, sports training titles, and interactive brand activations.
- Implemented real-time multiplayer with Photon, covering matchmaking, state synchronization and lag compensation across mobile and VR clients.
- Developed 2D/3D casual, hyper-casual and card games, plus high-performance WebGL titles delivered in-browser.
- Built native Android applications and integrated on-device machine-learning models for object detection.

Assistant Software Engineer
JUN 2021 – OCT 2021

Arclite Systems Ltd.
Dhaka, Bangladesh

- Developed 2D/3D casual and card games with replayable gameplay mechanics.
- Optimized builds for performance across mobile, desktop and VR platforms.

PUBLICATIONS

1. **I. Ahmed**, R. K. Kundu, and K. A. Hoque. "AegisVR: Leveraging Reinforcement Learning and LLMs for Personalized Cybersickness Mitigation and Reasoning in VR." *IEEE ISMAR*, 2026. (Accepted — ISMAR 2026)
2. **I. Ahmed**, R. K. Kundu, and K. A. Hoque. "Adversarial VR: An Open-Source Testbed for Evaluating Adversarial Robustness of VR Cybersickness Detection and Mitigation." *IEEE ISMAR-Adjunct*, pp. 281–288, 2025.
3. R. K. Kundu, **I. Ahmed**, and K. A. Hoque. "PrivateXR: Defending Privacy Attacks in Extended Reality Through Explainable AI-Guided Differential Privacy." *IEEE ISMAR*, pp. 507–517, 2025.
4. R. K. Kundu, **I. Ahmed**, and K. A. Hoque. "Enhancing Immersive Virtual Reality Experiences with Multiple Tasks Prediction Using Pre-Trained Large Foundation Models." *ACM VRST*, pp. 1–11, 2025.
5. R. K. Kundu, **I. Ahmed**, and K. A. Hoque. "Pilar: Personalizing Augmented Reality Interactions with LLM-Based Human-Centric and Trustworthy Explanations for Daily Use Cases." *IEEE ISMAR-Adjunct*, pp. 273–280, 2025.
6. R. K. Kundu, **I. Ahmed**, J. Quarles, and K. A. Hoque. "ExciteVR: Effective Cybersickness Mitigation in Virtual Reality using Explainable Artificial Intelligence and Large Language Models." *IEEE ISMAR-Adjunct*, pp. 644–647, 2025.

PROFESSIONAL SERVICE

- **Website Chair** and **Program Committee member**, TRUST-XR 2025 — International Workshop on Trustworthy, Secure and Privacy-Aware AI in Extended Reality, held in conjunction with *IEEE ISMAR 2025*, Daejeon, South Korea.
- **Reviewer**, *IEEE Transactions on Human-Machine Systems* (2 manuscripts).
- **Reviewer**, REALM Workshop, *EMNLP 2026* (1 manuscript).

AWARDS & MEDIA

- **Best Poster Award**, IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2025 — for *ExciteVR*.
- Featured in *Mizzou Engineering News* — Exposing the security and privacy risks posed by incorporating AI in XR.
- Featured in *Mizzou Engineering News* — Mizzou Engineering researchers bring home award from ISMAR.

EDUCATION

Ph.D. Computer Science — University of Missouri–Columbia
2025 – Present · Columbia, MO

M.S. Computer Science — University of Missouri–Columbia (expected December 2026)
2025 – Present · Columbia, MO

B.Sc. Computer Science & Engineering — Shahjalal University of Science and Technology
2016 – 2021 · Sylhet, BD